1 FIG-206 — Dft recovery reconstruction

Group: DFT-RECOVERY-RECONSTRUCTION **Evidence chain:** manifested **Status:** reconstruction-only diagnostic; source scope limited

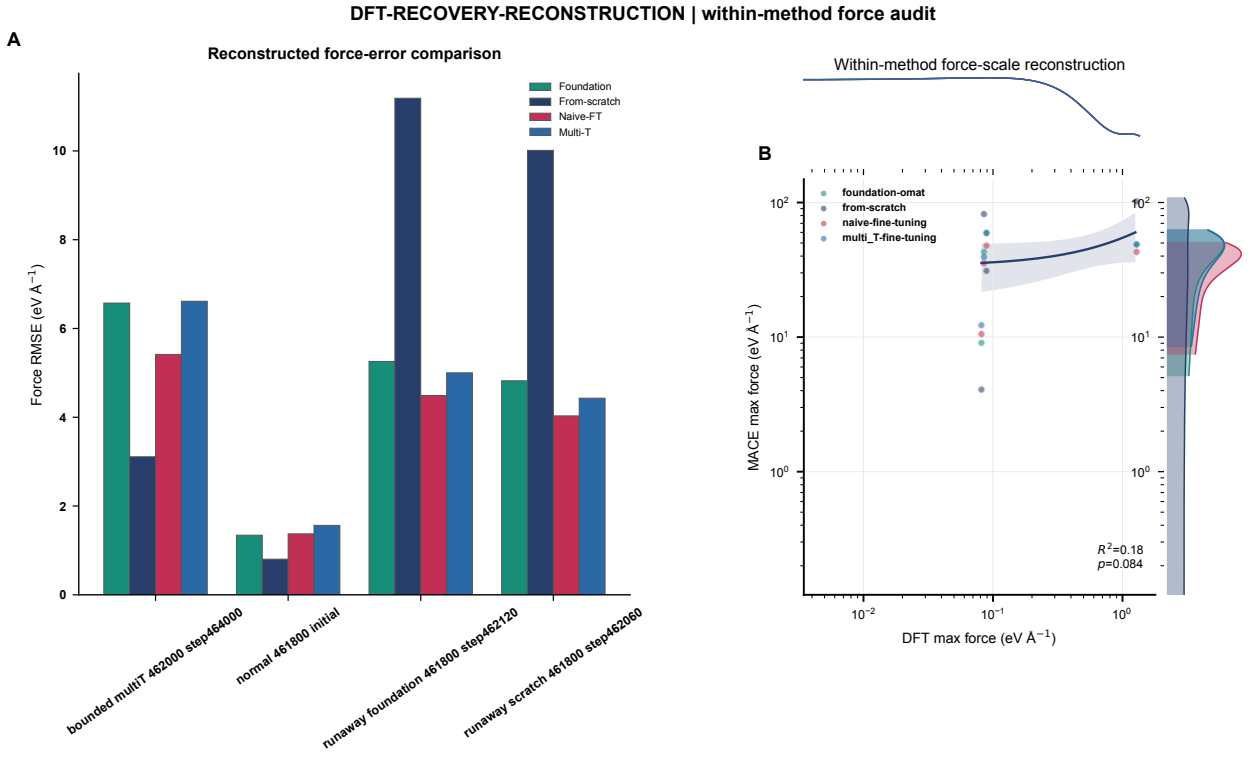


Figure 1: A, reconstructed DFT-versus-MACE force RMSE by recovery case and model; B, maximum-force relation for the same reconstructed rows. This page shows the Plot Atlas table-backed candidate "Dft recovery reconstruction"; its scientific status and caveats are reproduced from the package manifest.

Results text draft. The reconstruction table contains normal, bounded, and runaway cases with finite-looking DFT/MACE comparison fields, but every row is explicitly labeled RECONSTRUCTION / REPRODUCED_SCOPE_LIMITED in the source table. Model force errors increase strongly in the runaway cases. **Discussion and boundary.** This package is useful for auditing the DFT-recovery boundary and for deciding whether a later accepted QE release can replace the reconstruction. It must not be presented as a converged DFT validation benchmark. **Article placement.** Proposed article location: Results/SI: DFT recovery and MACE force-validation boundary. **Reviewer linkage:** overall status PARTIAL.

- **REV-A1 (PARTIAL)**: Reconcile Fig. 3d fixed-geometry DFT-path scoring with SI Fig. S3 self-consistent MLFF-NEB, including pathway identity and the DFT reference. *Completed:* The fixed-path versus self-relaxed operator distinction and the narrower model-reported-barrier wording are integrated. *Still required:* Close the restartable QE path-1–4 convergence gate, preserve the final path/source metadata, and complete the SI Fig. S3 endpoint/checkpoint/optimizer/source audit before calling the quantitative reference final.

Evidence pointers.

- Data/processed: `paper/analysis/phase3_dft_recovery_20261004/results_reconstructed/dft_vs_mace_within_method.csv`
- Data/processed: `paper/analysis/phase3_dft_recovery_20261004/results_reconstructed/qe_gate_audit.json`
- Plot/processing script: `main/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py`
- Caption/text source: `main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-RECOVERY-RECONSTRUCTION/caption_results_discussion.md`

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-RECOVERY-RECONSTRUCTION/caption_results_discussion.md
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/analysis/phase3_dft_recovery_20261004/results_reconstructed/dft_vs_mace_within_method.csv
- **Raw or processed data:** /Users/alina/Project/migrationbench-paper-repro/analysis/phase3_dft_recovery_20261004/results_reconstructed/qe_gate_audit.json
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-RECOVERY-RECONSTRUCTION/processed_dft_vs_mace_reconstruction.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- **Plot or processing script:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: The source scope is RECONSTRUCTION / REPRODUCED_SCOPE_LIMITED. A finite reconstructed row is not equivalent to a normal, accepted QE target. Re-run or replace this package after raw QE outputs and gate manifests are independently closed.